



NORTH SYDNEY BOYS HIGH SCHOOL

2009

Preliminary Examination

Mathematics

Teacher: Mr Berry

Instructions

- Working time – **2 hours 30 minutes**
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new **Question** is to be started on a **new page**.
- Attempt all questions

Student Name:

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	Total
Mark	$\frac{21}{21}$	$\frac{20}{20}$	$\frac{12}{12}$	$\frac{12}{12}$	$\frac{15}{15}$	$\frac{13}{13}$	$\frac{12}{12}$	$\frac{105}{105}$

Question 1 (21 Marks)		Use a Separate Sheet of paper	Marks
a)	Find the value of $\frac{1}{7.38} + \frac{1}{9.85}$, correct to 3 significant figures.		2
b)	Express the decimal $0.\dot{4}\dot{8}$ as a fraction in simplest form.		2
c)	If $\sqrt{56} + \sqrt{14} = \sqrt{A}$, find A .		2
d)	Express $\frac{4\sqrt{3}+3}{3\sqrt{3}-1}$ with a rational denominator. Simplify your answer.		2
e)	Factorise the following expressions fully:		
i)	$x^2 - 5x - 14$		1
ii)	$ax + ba + by + xy$		2
f)	Simplify: $(x-1)^2 - (x-2)^2$		2
g)	Simplify $\frac{10x-15}{6} \times \frac{1}{8x-12}$ as a single fraction in simplest form.		2
h)	Solve:		
i)	$6(y-1) = 3(y+8)$		2
ii)	$\frac{a+2}{3} = \frac{a}{2} - 2$		2
i)	Solve for x : $ 2x+2 < 8$.		2

End of Question 1.

Question 2 (20 Marks)

Use a Separate Sheet of paper

Marks

- a) If $f(x) = 2x^2 - x$ is this an odd function, even function or neither? **1**

- b) A function is defined by the rule $g(x) = \begin{cases} x+1, & \text{if } x \geq 1 \\ -1, & \text{if } -2 < x < 1 \\ 1-x, & \text{if } x \leq -2 \end{cases}$ **4**

Find if they exist,

- i) $g(1)$
- ii) $g(-1)$
- iii) $g(0)$
- iv) $g(2) + g(-2)$
- c) Sketch the graphs of the following, showing the x and y **intercepts**, stating the **domain** and **range** of each.
- i) $y = 2^x$ **4**
- ii) $x^2 + (y+3)^2 = 36$ **4**
- iii) $0 = 3x - y - 5$ **4**
- d) Show the region of the number plane where the following hold simultaneously: **3**
- $y \leq x+1$
- and $xy > 4$

End of Question 2.

Marks**Question 3 (12 Marks)**

Use a Separate Sheet of paper

- a) Find the derivative of the following: (You do not need to simplify your answers after finding the derivative.)

i) $2x^7 - 3x^5 + 5x^3 - 17$

2

ii) $\frac{1}{\sqrt[3]{x^4}}$

2

iii) $(x-2)(6x+7)$

2

iv) $\frac{2x^2 + 1}{5 - 3x^2}$

2

- b) Find $g'(-1)$ for $g(x) = (-x^4 + 3)^5$.

2

- c) Given $f(x) = (x+1)\sqrt{x}$ find $f'(x)$.

2**End of Question 3.**

Question 4 (12 Marks)		Use a Separate Sheet of paper	Marks
The points $A(2,0)$, $B(8,4)$, $C(4,6)$ and $D(x_1, y_1)$ form the 4 vertices of a parallelogram.			
a)	Draw a number plane and mark A , B & C on it.		1
b)	Find the gradient of line AB		1
c)	Show that the equation of the line l parallel to AB and going through C is $2x - 3y + 10 = 0$		2
d)	If the equation of the line k through A parallel to BC is $x + 2y - 2 = 0$. Find the point $D(x_1, y_1)$ the intersection of the lines l and k . Mark this point on your diagram.		2
e)	Find the angle θ to the nearest degree that the line AB makes with the positive x -axis		2
f)	Find the perpendicular distance between the line l and A .		2
g)	Find the exact area of $ABCD$		2

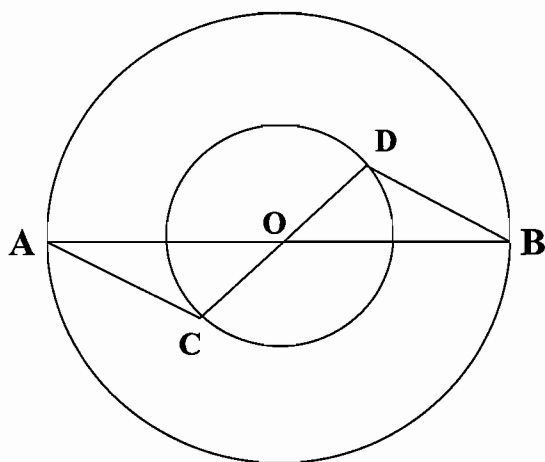
End of Question 4.

Question 5 (15 Marks)

Use a Separate Sheet of paper

Marks

- a) In the diagram below O is the centre of the two circles. AB is the diameter of the larger circle and CD is the diameter of the smaller circle.

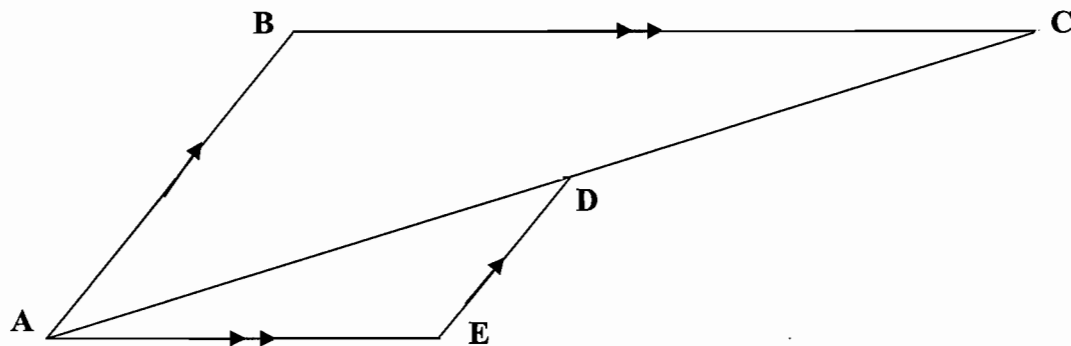


- | | | |
|-----|---|---|
| i) | Prove that $\triangle AOC \equiv \triangle DOB$ | 3 |
| | Hence, or otherwise, prove: | |
| ii) | $AC = DB$ | 1 |
| ii) | $AC \parallel DB$ | 1 |
- b) The sum of the interior angles of a regular polygon is 2700°
- | | | |
|------|---|---|
| i) | How many sides has the polygon? | 1 |
| ii) | Find the size of each interior angle to the nearest minute. | 1 |
| iii) | Hence find the size of each exterior angle. | 1 |

Question 5 continues on page 7

- c) Prove that $\triangle ABC$ is similar to $\triangle DEA$

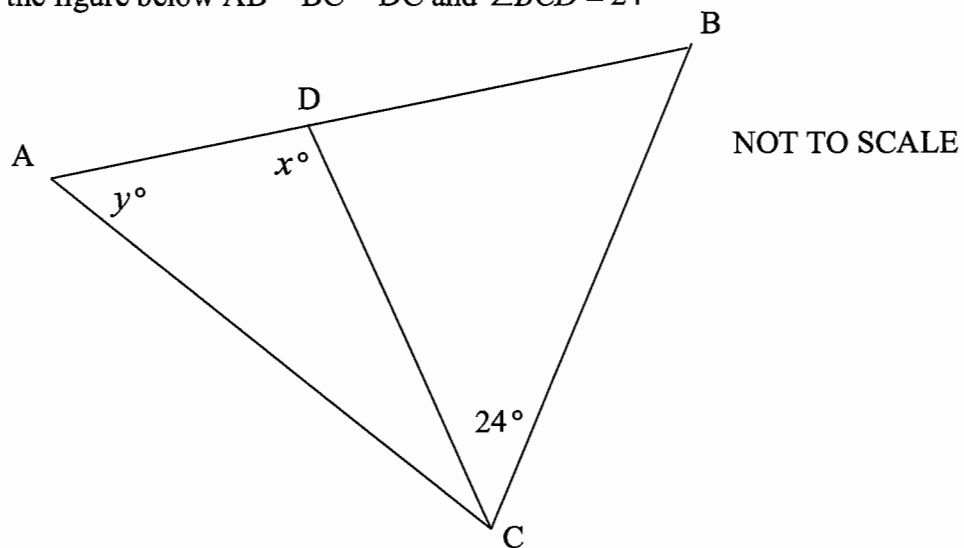
Marks



3

- d) In the figure below $AB = BC = DC$ and $\angle BCD = 24^\circ$

4



Find the values of x and y , giving reasons for each step.

End of Question 5.

Question 6 (13 Marks)	Use a Separate Sheet of paper	Marks
a) From a point 5m above the ground, the angle of depression of the bottom of a wall is 21° and the angle of elevation of the top of the wall is 32° .		
i) Draw a diagram to represent this information.		1
ii) Find the distance from the point of observation to the bottom of the wall. (correct to 2 decimal places)		2
iii) Using your answer from part (i) and the Sine Rule. Find the height of the wall. (correct to 2 decimal places)		2
b) Zoe and Kobi set out on a bike ride from point P at the same time. One travels at 20km/h along a straight road in the direction 032° T. The other travels at 25km/h along another straight road in the direction 132° T.		
ii) Draw a diagram to represent this information.		1
iii) Find the distance Zoe and Kobi are apart to the nearest kilometre after 3 hours.		2
c) Find the exact value of $\sec(60^\circ)$.		1
d) Solve $\sin \theta = \frac{-1}{\sqrt{2}}$ for $0^\circ \leq \theta \leq 360^\circ$.		2
e) Prove $\sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta}$.		2

End of Question 6.

Question 7 (12 Marks)*Start a new page***Marks**

(a) Solve for x : $(x + 1)^2 = 6$, leaving your answer in exact form. **2**

(b) Simplify $(4 - \sqrt{3})^3 - (4 + \sqrt{3})^3$ **3**

(c) Find the gradient of the curve $y = 2x^3 - 4x^2$ at the point $(1, -2)$ and hence find the equation of the normal to this curve at the point $(1, -2)$. **3**

(d) (i) Show that: $\tan\theta + \cot\theta = \frac{1}{\sin\theta \cos\theta}$ **1**

(ii) Hence or otherwise, solve: **3**

$$\frac{1 + \cot\theta}{\operatorname{cosec}\theta} - \frac{\sec\theta}{\tan\theta + \cot\theta} = -1, \quad 0^\circ \leq \theta \leq 360^\circ$$

End of Examination.

QUESTION 1

Solution	Marking Scheme
a) $\frac{1}{7.38} + \frac{1}{9.85} = 0.237 \text{ (3 s.f.)}$ b) $x = 0.\overline{48}$ $100x = 48.\overline{48}$ $100x - x = 48.\overline{48} - 0.\overline{48}$ $99x = 48$ $x = \frac{48}{99}$ $x = \frac{16}{33}$ c) $\begin{aligned}\sqrt{56} + \sqrt{14} &= \sqrt{4 \times 14} + \sqrt{14} \\ &= 2\sqrt{14} + \sqrt{14} \\ &= 3\sqrt{14} \\ &= \sqrt{9 \times 14} \\ &= \sqrt{126} \\ \therefore A &= 126\end{aligned}$ d) $\begin{aligned}&\frac{4\sqrt{3} + 3}{3\sqrt{3} - 1} \times \frac{3\sqrt{3} + 1}{3\sqrt{3} + 1} \\ &= \frac{(4\sqrt{3} + 3)(3\sqrt{3} + 1)}{(3\sqrt{3})^2 - 1^2} \\ &= \frac{12 \times 3 + 4\sqrt{3} + 9\sqrt{3} + 3}{27 - 1} \\ &= \frac{39 + 13\sqrt{3}}{26} \\ &= \frac{13(3 + \sqrt{3})}{2 \times 13} \\ &= \frac{3 + \sqrt{3}}{2}\end{aligned}$	

e) i)

$$x^2 - 5x - 14 = (x - 7)(x + 2)$$

ii)

$$\begin{aligned} ax + ba + by + xy &= a(x + b) + y(x + b) \\ &= (a + y)(x + b) \end{aligned}$$

f)

$$\begin{aligned} &(x - 1)^2 - (x - 2)^2 \\ &= ((x - 1) - (x - 2))((x - 1) + (x - 2)) \\ &= (1)((2x - 3)) \\ &= 2x - 3 \end{aligned}$$

g)

$$\begin{aligned} \frac{10x - 15}{6} \times \frac{1}{8x - 12} &= \frac{10x - 15}{6(8x - 12)} \\ &= \frac{5(2x - 3)}{24(2x - 3)} \\ &= \frac{5}{24} \end{aligned}$$

h) i)

$$\begin{aligned} 6(y - 1) &= 3(y + 8) \\ 6y - 6 &= 3y + 24 \\ 6y - 3y &= 24 + 6 \\ 3y &= 30 \\ \therefore y &= 10 \end{aligned}$$

ii)

$$\begin{aligned} \frac{a + 2}{3} &= \frac{a}{2} - 2 \\ \frac{a + 2}{3} &= \frac{a - 4}{2} \\ 2(a + 2) &= 3(a - 4) \\ 2a + 4 &= 3a - 12 \\ 3a - 2a &= 4 + 12 \\ \therefore a &= 16 \end{aligned}$$

i)

$$|2x + 2| < 8$$

Case 1:

$$(2x + 2) < 8$$

$$2x < 6$$

$$x < 3$$

Case 2:

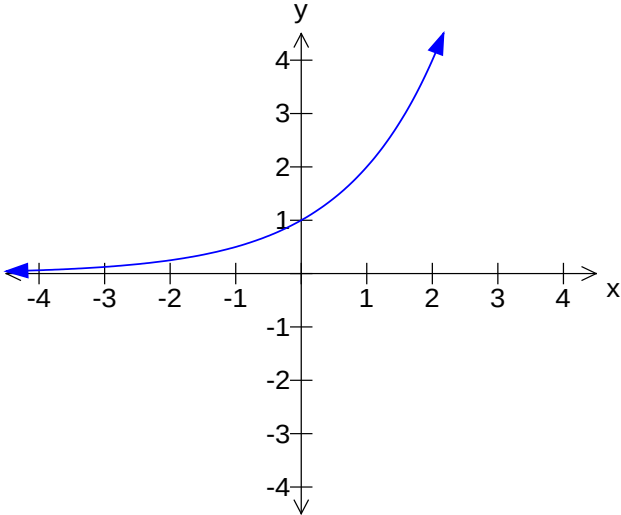
$$-(2x + 2) < 8$$

$$2x + 2 > -8$$

$$2x > -10$$

$$x > -5$$

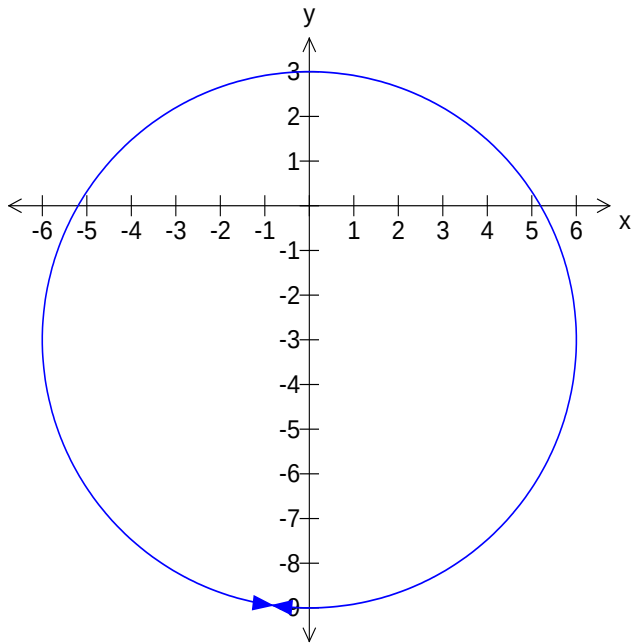
QUESTION 2

Solution	Marking Scheme
a) $f(x) = 2x^2 - x$ $f(-x) = 2(-x)^2 - (-x)$ $= 2x^2 + x$ $\neq \pm f(x)$ $\therefore$ neither even or odd b) i) $g(1) = 1 + 1$ $= 2$ ii) $g(-1) = -1$ iii) $g(0) = -1$ iv) $g(2) + g(-2) = (2 + 1) + (1 - (-2))$ $= 3 + 3$ $= 6$ c) 	

Domain: All real x

Range: $y > 0$

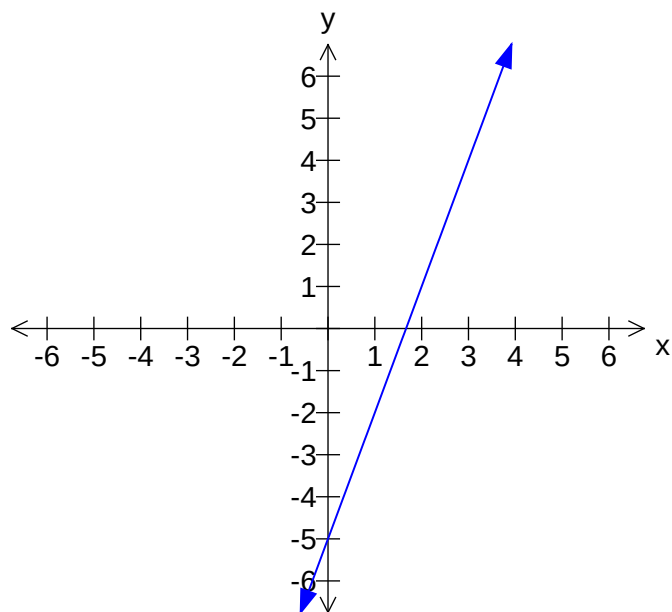
ii)



Domain: $-6 \leq x \leq 6$

Range : $-9 \leq y \leq 3$

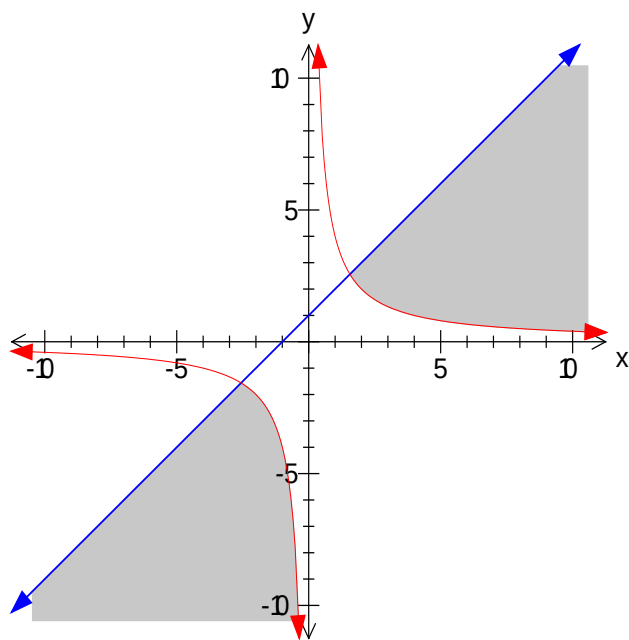
iii)



Domain: All real x

Range: All real y

d)



QUESTION 3

Solutions	Marking Scheme
a) i) $\frac{d}{dx} 2x^7 - 3x^5 + 5x^3 - 17$ $= 14x^6 - 15x^4 + 15x^2$ ii) $\frac{d}{dx} \left(\frac{1}{\sqrt[3]{x^4}} \right) = \frac{d}{dx} \left(x^{-\frac{4}{3}} \right)$ $= -\frac{4}{3} x^{-\frac{7}{3}}$ $= -\frac{4}{3(\sqrt[3]{x^7})}$ iii) $\frac{d}{dx} ((x-2)(6x+7))$ $= \left(\frac{d}{dx}(x-2) \right)(6x+7) + \left(\frac{d}{dx}(6x+7) \right)(x-2)$ $= (1)(6x+7) + (6)(x-2)$ $= 6x+7 + 6x-12$ $= 12x-5$ iv) $\frac{d}{dx} \left(\frac{2x^2+1}{5-3x^2} \right)$ $= \frac{\left(\frac{d}{dx} 2x^2+1 \right) (5-3x^2) - \left(\frac{d}{dx} 5-3x^2 \right) (2x^2+1)}{(5-3x^2)^2}$ $= \frac{(4x)(5-3x^2) - (-6x)(2x^2+1)}{(5-3x^2)^2}$ $= \frac{20x-12x^3+12x^3+6x}{(5-3x^2)^2}$ $= \frac{26x}{(5-3x^2)^2}$	

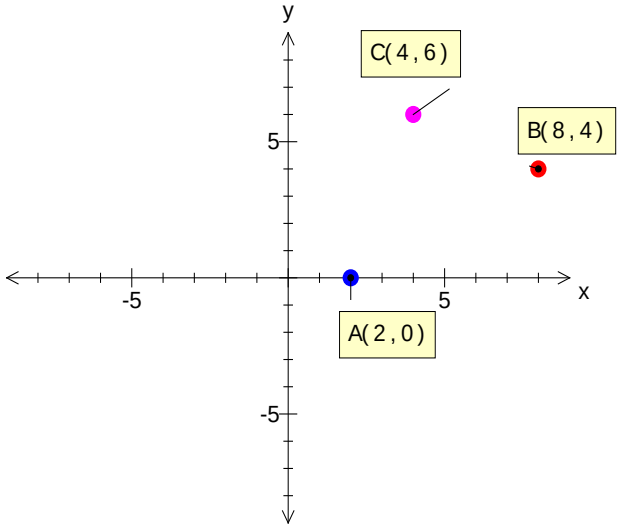
b)

$$\begin{aligned}g(x) &= (-x^4 + 3)^5 \\ \therefore g'(x) &= 5 \times (-x + 3)^4 \times -4x^3 \\ &= -20x^3(-x^4 + 3)^4 \\ \therefore g'(-1) &= -20(-1)^3(-(-1)^4 + 3)^4 \\ &= -20(4)^4 \\ &= -20 \times 64 \\ &= -1280\end{aligned}$$

c)

$$\begin{aligned}f(x) &= (x+1)\sqrt{x} \\ &= x\sqrt{x} + \sqrt{x} \\ &= x^{\frac{3}{2}} + x^{\frac{1}{2}} \\ f'(x) &= \frac{d}{dx}x^{\frac{3}{2}} + \frac{d}{dx}x^{\frac{1}{2}} \\ &= \frac{3}{2}x^{\frac{1}{2}} + \frac{1}{2}x^{-\frac{1}{2}} \\ &= \frac{3\sqrt{x}}{2} + \frac{1}{2\sqrt{x}} \\ &= \frac{3(\sqrt{x})^2}{2\sqrt{x}} + \frac{1}{2\sqrt{x}} \\ &= \frac{3x+1}{2\sqrt{x}} \\ &= \frac{3x+1}{2\sqrt{x}} \times \frac{\sqrt{x}}{\sqrt{x}} \\ &= \frac{\sqrt{x}(3x+1)}{2x}\end{aligned}$$

QUESTION 4

Solutions	Marking Scheme
a)  b) $m_{AB} = \frac{y_B - y_A}{x_B - x_A}$ $= \frac{4 - 0}{8 - 2}$ $= \frac{4}{6}$ $= \frac{2}{3}$ c) $y - y_1 = m(x - x_1)$ $y - 6 = \frac{2}{3}(x - 4)$ $3y - 18 = 2x - 8$ $3y - 2x - 18 + 8 = 0$ $\therefore 2x - 3y + 10 = 0$ as required	

d) $2x - 3y + 10 = 0$ ①

$x + 2y - 2 = 0$ ②

$2 \times$ ②

$2x + 4y - 4 = 0$ ③

① - ③

$-7y + 14 = 0$

$7y = 14$

$\therefore y = 2$

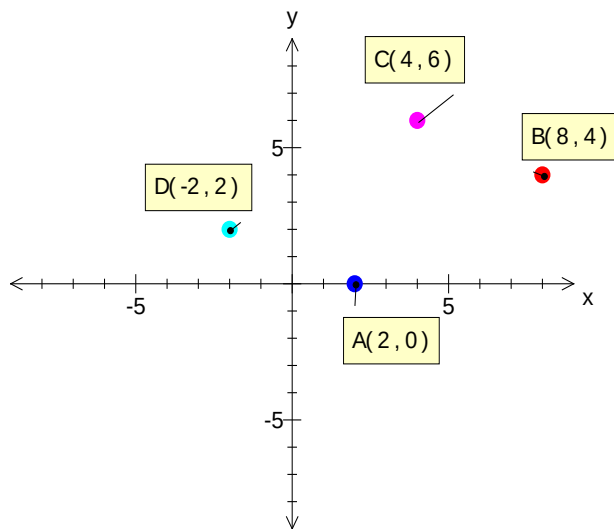
substitute into ②

$x + (2 \times 2) - 2 = 0$

$x + 2 = 0$

$x = -2$

$\therefore D = (-2, 2)$



e)

$\tan \theta = \frac{\text{rise}}{\text{run}}$

$= \frac{4 - 0}{8 - 2}$

$\therefore \theta = \tan^{-1}\left(\frac{4}{6}\right)$

$= 34^\circ$ (to the nearest degree)

f)

$$\begin{aligned}d &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\&= \frac{|(2)(2) + (-3)(0) + (10)|}{\sqrt{2^2 + (-3)^2}} \\&= \frac{|4 + 10|}{\sqrt{13}} \\&= \frac{14}{\sqrt{13}} \text{ units}\end{aligned}$$

g)

$$\begin{aligned}D_{AB} &= \sqrt{((8 - 2)^2 + (4 - 0)^2)} \\&= \sqrt{52} \\&= \sqrt{4 \times 13} \\&= 2\sqrt{13} \\A &= bh \\&= 2\sqrt{13} \times \frac{14}{\sqrt{13}} \\&= 28 \text{ units}^2\end{aligned}$$

QUESTION 5

Solutions	Marking Scheme
a) i) <i>In Δs AOC & ODB</i> <i>AO = BO (radii of large circle)</i> <i>CO = DO (radii of small circle)</i> <i>$\angle AOC = \angle BOD$ (vertically opposite)</i> <i>$\therefore \Delta AOC = \Delta ODB$ (SAS)</i> ii) <i>AC = DB (matching sides in congruent triangles AOC and ODB)</i> iii) <i>$\angle ACO = \angle BDO$ (matching angles in congruent triangles)</i> <i>AC $\parallel$ DB (alternate angles ACO & BDO are equal)</i> b) i) $(n - 2) \times 180^\circ = 2700$ $n - 2 = 15$ $n = 17$ <i>$\therefore$ 17 sides</i> ii) $\angle = 2700^\circ \div 17$ $= 158^\circ 49' \text{ (nearest minute)}$ iii) $\angle = 180^\circ - 158^\circ 49'$ $= 21^\circ 11' \text{ (nearest minute)}$ c) <i>In Δs ABC & DEA</i> <i>$\angle DAE = \angle ACB$ (alternate angles, AE $\parallel$ BC)</i> <i>$\angle BAC = \angle ADE$ (alternate angles, AB $\parallel$ DE)</i> <i>$\therefore \Delta ABC \parallel \Delta DEA$ (equi-angular)</i>	

d)

$$2\angle CDE + 24^\circ = 180^\circ \text{ (angle sum of } \triangle BCD)$$

$$2\angle CDE = 156^\circ$$

$$\therefore \angle CDE = 78^\circ$$

$$\therefore x = 102^\circ \text{ (on a straight line with } \angle CDE)$$

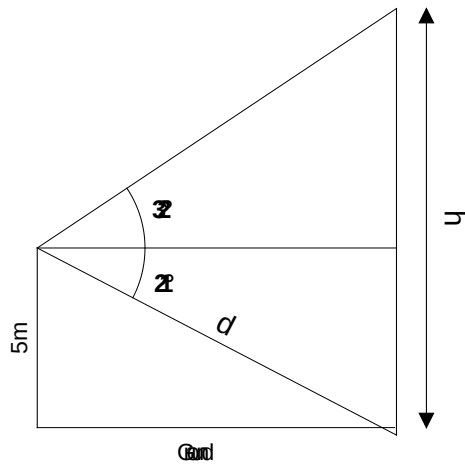
$$2y^\circ + 78^\circ = 180^\circ$$

$$2y^\circ = 102^\circ$$

$$\therefore y = 51^\circ$$

QUESTION 6

a) i)



ii)

$$\sin(21^\circ) = \frac{5}{d}$$

$$d = \frac{5}{\sin(21^\circ)}$$

$$\therefore d = 13.95m \text{ (2 d.p.)}$$

iii)

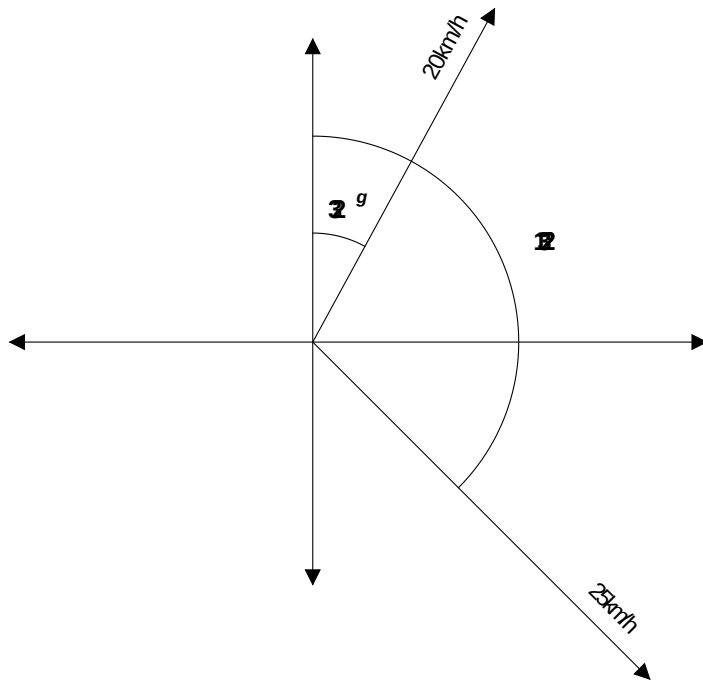
$$\frac{h}{\sin(53^\circ)} = \frac{d}{\sin(90^\circ - 32^\circ)}$$

$$h \sin(58^\circ) = d \sin(53^\circ)$$

$$h = \frac{d \times \sin(53^\circ)}{\sin(58^\circ)}$$

$$\therefore h = 13.14m \text{ (2 d.p.)}$$

b)



let the distance be d

$$d^2 = (20 \times 3)^2 + (25 \times 3)^2 - 2(20 \times 3)(25 \times 3)\cos(132 - 32)$$

$$d = \sqrt{60^2 + 75^2 - (2 \times 60 \times 75 \times \cos(100))}$$

$$\therefore d = 104 \text{ km (nearest km)}$$

c)

$$\begin{aligned} \sec(60^\circ) &= \frac{1}{\cos(60^\circ)} \\ &= \frac{1}{\left(\frac{1}{2}\right)} \\ &= 2 \end{aligned}$$

d)

$$\sin\theta = \frac{-1}{\sqrt{2}}$$

$$\theta = \sin^{-1}\left(\frac{-1}{\sqrt{2}}\right)$$

$$\therefore \theta = 225^\circ \text{ or } 315^\circ$$

e)

$$\begin{aligned}\sec\theta + \tan\theta &= \frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta} \\ &= \frac{1 + \sin\theta}{\cos\theta}\end{aligned}$$

as required

QUESTION 7

Solutions

a)

$$(x + 1)^2 = 6$$

$$x + 1 = \pm\sqrt{6}$$

$$x = -1 \pm \sqrt{6}$$

b)

$$\begin{aligned} & (4 - \sqrt{3})^3 - (4 + \sqrt{3})^3 \\ &= ((4 - \sqrt{3}) - (4 + \sqrt{3}))((4 - \sqrt{3})^2 + (4 + \sqrt{3})(4 - \sqrt{3}) + (4 + \sqrt{3})^2) \\ &= (-2\sqrt{3})((16 - 8\sqrt{3} + 3) + (16 - 3) + (16 + 8\sqrt{3} + 3)) \\ &= (-2\sqrt{3})(51) \\ &= -102\sqrt{3} \end{aligned}$$

c)

$$\frac{dy}{dx} = 6x^2 - 4x$$

$$\text{at } x = 1$$

$$\begin{aligned} m_{TANGENT} &= 6(1)^2 - 4(1) \\ &= 2 \end{aligned}$$

$$\therefore m_{NORMAL} = -\frac{1}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - (-2) = -\frac{1}{2}(x - 1)$$

$$-2(y + 2) = x - 1$$

$$-2y - 4 = x - 1$$

$$\therefore x + 2y + 3 = 0$$

d) i)

$$\begin{aligned} \tan\theta + \cot\theta &= \frac{\sin\theta}{\cos\theta} + \frac{1}{\tan\theta} \\ &= \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} \\ &= \frac{\sin\theta\sin\theta + \cos\theta\cos\theta}{\sin\theta\cos\theta} \\ &= \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} \\ &= \frac{1}{\sin\theta\cos\theta} \end{aligned}$$

as required

ii)

$$\frac{1 + \cot\theta}{\operatorname{cosec}\theta} - \frac{\sec\theta}{\tan\theta + \cot\theta} = -1$$

$$\frac{1 + \frac{\cos\theta}{\sin\theta}}{\frac{1}{\sin\theta}} - \frac{\frac{1}{\cos\theta}}{\frac{1}{\sin\theta\cos\theta}} = -1$$

$$\sin\theta\left(1 + \frac{\cos\theta}{\sin\theta}\right) - \sin\theta\cos\theta\left(\frac{1}{\cos\theta}\right) = -1$$

$$\sin\theta + \cos\theta - \sin\theta = -1$$

$$\cos\theta = -1$$

$$\theta = \cos^{-1}(-1)$$

$\therefore$

$$\theta = 180^\circ$$